Submission Checklist

NeuroSonic project · Allwin Raj R (26MAI0072), Madhu Sudharshanan (26MAI0071) · Vellore Institute of Technology · November 2026

Live demo video: https://drive.google.com/file/d/1yJS2f8_ee3ZwV55mkA9e1c6mm9f5xkYd/view?usp=sharing

Submission Package: Contents and Mapping

Project: **NeuroSonic** (base paper: WOAD, Gao et al., *Nature Communications* 2025). Students: Allwin Raj R (26MAI0072) and Madhu Sudharshanan (26MAI0071), M.Tech. CSE (AI and ML); course MACSE517N Computer Architecture and Organisation; guide: Yogesh Babu S. Prepared 3 November 2026. All files are in the `submission/` folder of the project repository, numbered in the order of the submission list.

Required item	File	Pages	Notes
Review 1 doc / DA1: survey and comparative analysis (IEEE 2-column, ≥ 7 pages)	`01_DA1_Review1_Survey_and_Comparative_Analysis.pdf`	9	domain; statistics; publications; patents; products; technology trends
Review 2 doc / DA2: base paper redo, gaps, proposed work (IEEE 2-column, ≥ 7 pages)	`02_DA2_Review2_Base_Paper_Reimplementation_Gaps_and_Proposed_Work.pdf`	7	re-implementation, evaluation, cons, 8 gaps, proposed work
Review 3 doc / DA3: implementation, architecture/methodology, results (IEEE 2-column, ≥ 7 pages)	`03_DA3_Review3_Implementation_Architecture_and_Results.pdf`	9	architecture, methods, all measured results, limitations
Final consolidated project report (≥ 20 pages; one upload for DA1–DA3)	`05_Final_Consolidated_Project_Report.pdf`	65	final edited version: all chapters, informal iPhone screenshots, appendices
Lab 9: IPR IDF B draft	`09_Lab9_IPR_IDF_B_Draft.pdf`	8	transfer to the institute template if one is provided
Lab 10 / IEEE 2-column draft (≥ 14 pages)	`06_Lab10_IEEE_Two_Column_Paper_Draft.pdf`	20	full paper draft (also serves as "IEEE 2-column draft of minimum 7 pages")
Project demo: existing implementation vs proposed work (differentiation)	`04_Project_Demo_Existing_vs_Proposed.pdf`	2	A/B/C tables and the demonstration script
GitHub link (dataset, source code, other files); video link	`07_GitHub_Dataset_and_Video_Links.pdf`	3	GitHub and video links; repository contents; video shot plan
Identification of Scopus-indexed venues (Q1–Q4) / book chapter / IPR	`08_Publication_Venue_Identification.pdf`	3	candidate venues, quartiles as checked, plan

To complete before submission

- Complete the remaining highlighted fields in the IDF: contribution percentages, Madhu Sudharshanan's e-mail, applicant, funding, planned publication date, signature date.
- The GitHub repository (with the demonstration video) is private until the IPR cell has given advice; add the faculty as collaborators to give them access.
- Physical iPhone tests are **not** part of these documents' results; every document says so.

How the documents were produced

All documents are generated from HTML sources in `_build/` (`python build.py`, headless Chrome to PDF). New figures come from `_build/make_figures.py`, which reads the project's result files; the other figures come from `publication/figures/`. Every cited source was checked on 3 November 2026; every measured number comes from files in `outputs/`.